

Lin Wang 王林

Postdoctoral Researcher | Computational Molecular Discovery

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I develop computational methods that integrate chemical priors into molecular representation learning and generative drug design. My research spans AI-driven drug discovery and protein–ligand and protein–protein interaction modeling, with emerging directions in phenotype-based and polypharmacology drug design. My latest work, Ouroboros, advances this broader program through a molecular foundation model that learns chemically informed representations from conformational-space and pharmacophore similarities for molecular property prediction and generation.

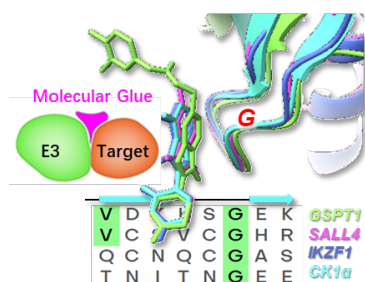
Research Focus

- Molecular representation and generation foundation models for medicinal chemistry
- Conformational-space- and pharmacophore-aware molecular representations
- Protein–ligand and protein–protein interaction modeling
- Phenotype-based and polypharmacology drug design

Education and Experience

Current	Postdoctoral Researcher Institute of Systems Medicine, Chinese Academy of Medical Sciences, Suzhou, China.
2019-2024	Ph.D., ShanghaiTech University
2019	B.S. in Life Science, Northeast Agricultural University

Selected Research



2022 | PPI and molecule glue

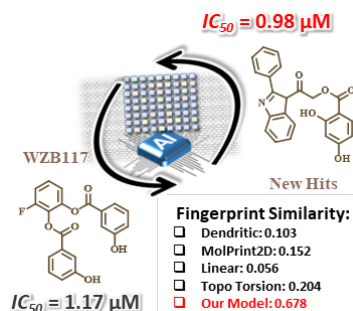
PPI-Miner

A motif-driven workflow that searches the proteome primarily by backbone-structure similarity, with sequence similarity as complementary evidence. Matched motifs are aligned to transfer a binding mode, then assembled and optimized as candidate protein-protein complexes.

Highlights

- Retrieves sequence-divergent proteins whose motifs preserve the query's backbone geometry, including beta-hairpins
- Proteome-wide CRBN mining predicted 1,739 candidate substrates; 16 had been reported experimentally before this study

[Paper](#) | [Code](#) | [CRBN library](#)



2024 | Molecular representation

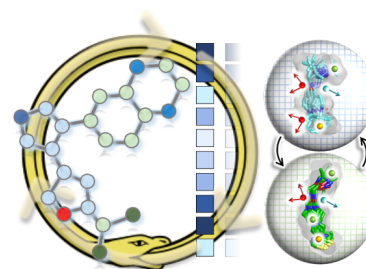
GeminiMol

A pairwise contrastive framework that trains a graph encoder to predict conformational-space and pharmacophore similarities. The learned encoding transfers to molecular-property and cellular-activity prediction, virtual screening, target identification, and scaffold hopping.

Highlights

- Screening 18 million compounds identified the scaffold-distinct GM-10, validated by whole-cell patch clamp against GluN1/GluN3A ($IC_{50} = 0.98 \pm 0.13 \mu M$)
- First prize in the 2023 Shanghai International Computational Biology Innovation Competition

[Paper](#) | [Code](#) | [Application](#)



2026 | Representation & generation

Ouroboros

A molecular foundation model that regularizes graph encodings with fingerprint, conformational-space, and pharmacophore similarity matrices. Its unified latent space supports molecular property prediction, similarity-based screening, targeted polypharmacology, and directed molecular optimization.

Highlights

- Full similarity-matrix learning integrates conformational-space and pharmacophore similarities into a shared molecular representation
- A shared representation connects molecular property prediction with property-guided generation

[Paper](#) | [Code](#)

Publications

= co-first author; * = co-corresponding author.

2026

Integrating chemical priors and physical laws to mitigate hallucinations in structure-based drug design

Zhongyu Liu, Yadong Liu[#], Yihang Zhou, Zhiyuan Liu, Yuhang Yang, Zhiwen Wang, Tianyi Zang^{*}, **Lin Wang**[#], Yang Zhang
Communications Chemistry, 2026 - [Google Scholar](#)

Learned Conformational Space and Pharmacophore Into Molecular Foundational Model

Lin Wang, Yifan Wu, Hao Luo, Minglong Liang, Yihang Zhou, Cheng Chen, Chris Liu, Jun Zhang, Yang Zhang
Advanced Science, p. e13556, 2026 - [DOI](#)

Molecular glue degraders of HuR suppress BRAF-mutant colorectal cancer

Xiaocui Lu, Xiuyun Wang, Zheng Yang, Xusheng Wang, Lin Wang, Chunhui Xu, I Lo, Chenlu Geng, Yisheng Pu, Keyu Zhang, et al.
Nature, p. 1–10, 2026 - [Google Scholar](#)

ProteinConformers: Benchmark Dataset for Simulating Protein Conformational Landscape Diversity and Plausibility

Yihang Zhou, Chen Wei, Minghao Sun, Jin Song, Yang Li, Lin Wang, Yang Zhang
Advances in Neural Information Processing Systems, vol. 38, 2026 - [Google Scholar](#)

Small Molecule Conformation Prediction

Lin Wang, Fang Bai
Artificial Intelligence for Drug Design, p. 527–546, book chapter, 2026 - [Google Scholar](#)

2025

DeepPSA: A Geometric Deep Learning Model for PROTAC Synthetic Accessibility Prediction

Ran Zhang, Shihang Wang, Lin Wang, Siyuan Tian, Yilin Tang, Fang Bai
Journal of Chemical Information and Modeling, vol. 65(13), p. 6861–6873, 2025 - [Google Scholar](#)

Discovery of novel GluN1/GluN3A NMDA receptor inhibitors using a deep learning-based method

Shihang Wang[#], Yue Zeng[#], Hao Yang[#], Si-yuan Tian[#], Yong-qi Zhou[#], **Lin Wang**[#], Xue-qin Chen, Hai-ying Wang, Zhao-bing Gao^{*}, Fang Bai
Acta Pharmacologica Sinica, vol. 46(11), p. 3108–3115, 2025 - [Google Scholar](#)

Fitness costs of mobilised colistin resistance gene 3 (mcr-3): systematic review, epidemiological study, and functional analysis

Lujie Liang[#], Yaxin Li[#], **Lin Wang**[#], Wenli Wang, Yihao Zhang, Hui Zhao, Yaxuan Wang, Lingxuan Lyu, Jiachen Li, Dianrong Zhou, Zhe Hu, Lizhen Luo, Guanxiu Wang, Jia Wan, Lin Xu, Meisong Li, Min Dai, Meiting Yang, Shun Xiong, Lan-Lan Zhong, Fang Bai, Siyuan Feng, Guo-Bao Tian
EBioMedicine, vol. 120, p. 105923, 2025 - [DOI](#)

MAI-TargetFisher: A proteome-wide drug target prediction method synergetically enhanced by artificial intelligence and physical modeling

Shi-wei Li, Peng-xuan Ren, Lin Wang, Qi-lei Han, Feng-lei Li, Hong-lin Li, Fang Bai
Acta Pharmacologica Sinica, vol. 46(5), p. 1462–1475, 2025 - [Google Scholar](#)

NADPH-Independent Fluorescent Probe for Live-Cell Imaging of Heme Oxygenase-1

Liang Li, Xuanyi Lu, Qiyan He, Chao Shu, Edward RH Walter^{*}, **Lin Wang**^{*}, Nicholas J Long^{*}, Lijun Jiang^{*}
ACS sensors, vol. 10(1), p. 499–506, 2025 - [Google Scholar](#)

PhenoModel: A multimodal phenotypic drug design foundation model for discovering novel potential inhibitors of multiple cancer cells

Shihang Wang[#], Qilei Han[#], Weichen Qin[#], **Lin Wang**[#], Junhong Yuan, Fengyu Cai, Yiqun Zhao, Pengxuan Ren, Yunze Zhang, Yilin Tang, et al.
Acta Pharmaceutica Sinica B, 2025 - [DOI](#)

Recent advances in molecular representation methods and their applications in scaffold hopping

Shihang Wang, Ran Zhang, Xiangcheng Li, Fengyu Cai, Xinyue Ma, Yilin Tang, Chao Xu, Lin Wang, Pengxuan Ren, Lu Liu, et al.
Npj Drug Discovery, vol. 2(1), p. 14, 2025 - [Google Scholar](#)

2024

Conformational space profiling enhances generic molecular representation for AI-powered ligand-based drug discovery

Lin Wang, Shihang Wang, Hao Yang, Shiwei Li, Xinyu Wang, Yongqi Zhou, Siyuan Tian, Lu Liu, Fang Bai
Advanced Science, vol. 11(40), p. 2403998, 2024 - [DOI](#)

Salvianolic acid B attenuates liver fibrosis by targeting Ecm1 and inhibiting hepatocyte ferroptosis

Yadong Fu, Xiaoxi Zhou, Lin Wang, Weiguo Fan, Siqi Gao, Danyan Zhang, Zhiyang Ling, Yaguang Zhang, Liyan Ma, Fang Bai, et al.
Redox biology, vol. 69, p. 103029, 2024 - [Google Scholar](#)

2023

A new variant of the colistin resistance gene MCR-1 with co-resistance to β -lactam antibiotics reveals a potential novel antimicrobial peptide

Lujie Liang[#], Lan-Lan Zhong[#], **Lin Wang**[#], Dianrong Zhou, Yaxin Li, Jiachen Li, Yong Chen, Wanfei Liang, Wenjing Wei, Chenchen Zhang, et al.
PLoS Biology, vol. 21(12), p. e3002433, 2023 - [Google Scholar](#)

DeepSA: a deep-learning driven predictor of compound synthesis accessibility

Shihang Wang[#], **Lin Wang**[#], Fenglei Li, Fang Bai
Journal of Cheminformatics, vol. 15(1), p. 103, 2023 - [DOI](#)

Discovery, evaluation and mechanism study of WDR5-targeted small molecular inhibitors for neuroblastoma

Qi-lei Han, Xiang-lei Zhang, Peng-xuan Ren, Liang-he Mei, Wei-hong Lin, Lin Wang, Yu Cao, Kai Li^{*}, Fang Bai^{*}
Acta Pharmacologica Sinica, vol. 44(4), p. 877–887, 2023 - [Google Scholar](#)

2023 (continued)

Discovery, structural optimization, and anti-tumor bioactivity evaluations of betulinic acid derivatives as a new type of ROR γ antagonists

Lianghe Mei, Lansong Xu, Sanan Wu, Yafang Wang, Chao Xu, Lin Wang, Xingyu Zhang^{*}, Chengcheng Yu, Hualiang Jiang, Xianglei Zhang^{*}, et al.

European Journal of Medicinal Chemistry, vol. 257, p. 115472, 2023 - [Google Scholar](#)

Propofol exerts anti-anhedonia effects via inhibiting the dopamine transporter

Xiao-Na Zhu, Jie Li, Gao-Lin Qiu, Lin Wang, Chen Lu, Yi-Ge Guo, Ke-Xin Yang, Fang Cai, Tao Xu^{*}, Ti-Fei Yuan^{*}, et al.

Neuron, vol. 111(10), p. 1626–1636, 2023 - [Google Scholar](#)

Synthesis of maleimide-braced peptide macrocycles and their potential anti-SARS-CoV-2 mechanisms

Jian Li, Jina Sun, Xianglei Zhang, Ruxue Zhang, Qian Wang, Lin Wang, Leike Zhang, Xiong Xie, Chunpu Li, Yu Zhou, et al.

Chemical Communications, vol. 59(7), p. 868–871, 2023 - [Google Scholar](#)

2022

A multi-targeting drug design strategy for identifying potent anti-SARS-CoV-2 inhibitors

Peng-xuan Ren, Wei-juan Shang, Wan-chao Yin, Huan Ge, Lin Wang, Xiang-lei Zhang, Bing-qian Li, Hong-lin Li, Ye-chun Xu, Eric H Xu, et al.

Acta Pharmacologica Sinica, vol. 43(2), p. 483–493, 2022 - [Google Scholar](#)

Crystal structures of Wolbachia CidA and CidB reveal determinants of bacteria-induced cytoplasmic incompatibility and rescue

Haofeng Wang, Yunjie Xiao, Xia Chen, Mengwen Zhang, Guangxin Sun, Feng Wang, Lin Wang, Hanxiao Zhang, Xiaoyu Zhang, Xin Yang, et al.

Nature Communications, vol. 13(1), p. 1608, 2022 - [Google Scholar](#)

Discovery of potential small molecular SARS-CoV-2 entry blockers targeting the spike protein

Lin Wang[#], Yan Wu[#], Sheng Yao[#], Huan Ge[#], Ya Zhu, Kun Chen, Wen-zhang Chen, Yi Zhang, Wei Zhu^{*}, Hong-yang Wang, et al.

Acta Pharmacologica Sinica, vol. 43(4), p. 788–796, 2022 - [Google Scholar](#)

PiLSL: pairwise interaction learning-based graph neural network for synthetic lethality prediction in human cancers

Xin Liu, Jiale Yu, Siyu Tao, Beiyuan Yang, Shike Wang, Lin Wang, Fang Bai, Jie Zheng

Bioinformatics, vol. 38(Supplement_2), p. ii106–ii112, 2022 - [Google Scholar](#)

PPI-Miner: a structure and sequence motif co-driven protein–protein interaction mining and modeling computational method

Lin Wang, Feng-lei Li, Xin-yue Ma, Yong Cang, Fang Bai

Journal of Chemical Information and Modeling, vol. 62(23), p. 6160–6171, 2022 - [DOI](#)

Rapamycin targets STAT3 and impacts c-Myc to suppress tumor growth

Le Sun, Yu Yan, Heng Lv, Jianlong Li, Zhiyuan Wang, Kun Wang, Lin Wang, Yunxia Li, Hong Jiang, Yaoyang Zhang

Cell chemical biology, vol. 29(3), p. 373–385, 2022 - [Google Scholar](#)

Recent advances in predicting protein–protein interactions with the aid of artificial intelligence algorithms

Shiwei Li, Sanan Wu, Lin Wang, Fenglei Li, Hualiang Jiang, Fang Bai

Current Opinion in Structural Biology, vol. 73, p. 102344, 2022 - [Google Scholar](#)

Target-based virtual screening and LC/MS-guided isolation procedure for identifying phloroglucinol-terpenoid inhibitors of SARS-CoV-2

Bo Hou, Yu-Min Zhang, Han-Yi Liao, Li-Feng Fu, De-Dong Li, Xin Zhao, Jian-Xun Qi, Wei Yang, Geng-Fu Xiao, Lian Yang, et al.

Journal of natural products, vol. 85(2), p. 327–336, 2022 - [Google Scholar](#)

2021

CKB inhibits epithelial-mesenchymal transition and prostate cancer progression by sequestering and inhibiting AKT activation

Zheng Wang, Mohit Hulsurkar, Lijuan Zhuo, Jinbang Xu, Han Yang, Samira Naderinezhad, Lin Wang, Guoliang Zhang, Nanping Ai, Linna Li, et al.

Neoplasia, vol. 23(11), p. 1147–1165, 2021 - [Google Scholar](#)

Identification of potential binding sites of sialic acids on the RBD domain of SARS-CoV-2 spike protein

Bingqian Li, Lin Wang, Huan Ge, Xianglei Zhang, Penxuan Ren, Yu Guo, Wuyan Chen, Jie Li, Wei Zhu, Wenzhang Chen, et al.

Frontiers in Chemistry, vol. 9, p. 659764, 2021 - [Google Scholar](#)

Probing the allosteric inhibition mechanism of a spike protein using molecular dynamics simulations and active compound identifications

Qian Wang[#], Lin Wang[#], Yumin Zhang[#], XiangLei Zhang, Leike Zhang, Weijuan Shang, Fang Bai^{*}

Journal of Medicinal Chemistry, vol. 65(4), p. 2827–2835, 2021 - [Google Scholar](#)

2020

Antimicrobial resistance in clinical *Ureaplasma* spp. and *Mycoplasma hominis* and structural mechanisms underlying quinolone resistance

Ting Yang, Lianlian Pan, Ningning Wu, Lin Wang, Zhen Liu, Yingying Kong, Zhi Ruan, Xinyou Xie, Jun Zhang

Antimicrobial Agents and Chemotherapy, vol. 64(6), p. 10–1128, 2020 - [Google Scholar](#)

Structure of Mpro from SARS-CoV-2 and discovery of its inhibitors

Zhenming Jin, Xiaoyu Du, Yechun Xu, Yongqiang Deng, Meiqin Liu, Yao Zhao, Bing Zhang, Xiaofeng Li, Leike Zhang, Chao Peng, et al.

Nature, vol. 582(7811), p. 289–293, 2020 - [DOI](#)

The structural study of mutation-induced inactivation of human muscarinic receptor M4

Jingjing Wang, Meng Wu, Lijie Wu, Yueming Xu, Fei Li, Yiran Wu, Petr Popov, Lin Wang, Fang Bai, Suwen Zhao, et al.

IUCrJ, vol. 7(2), p. 294–305, 2020 - [Google Scholar](#)